Status	Finished
Started	Saturday, 21 March 2026, 4:07 PM
Completed	Saturday, 21 March 2026, 4:22 PM
Duration	14 mins 32 secs
Grade	**30.00** out of 40.00 (**75**%)

Question 1

Correct

Mark 1.00 out of 1.00

```
What will be the output?

import numpy as np

a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
print(np.dot(a, b))
```

Select one:

- a. Error
- b. 32 ✓
- c. [4, 10, 18]
- d. 15

The correct answer is: 32

Question 2

Correct

Mark 1.00 out of 1.00

```
What is the output type of this code?

import plotly.express as px
import pandas as pd

df = pd.DataFrame({
    'year': [2020, 2021, 2022],
    'sales': [100, 150, 200]
})

fig = px.line(df, x='year', y='sales')
fig.show()
```

Select one:

- a. Bar chart
- b. Interactive line chart
- c. Static text output
- d. Heatmap

The correct answer is: Interactive line chart

Question 3

Incorrect

Mark 0.00 out of 1.00

```
What will this code do?

import sqlite3

con = sqlite3.connect(':memory:')
cur = con.cursor()
cur.execute("CREATE TABLE test(id INTEGER, name TEXT)")
cur.execute("INSERT INTO test VALUES(1, 'A')")
cur.execute("SELECT * FROM test")
print(cur.fetchall())
```

Select one:

- a. [(1, 'A')]
- b. Empty list
- c. [1, 'A']
- d. Error

The correct answer is: [(1, 'A')]

Question 4

Correct

Mark 1.00 out of 1.00

```
What will be printed?

class MyError(Exception):
    pass

try:
    raise MyError("Custom")
except Exception as e:
    print(type(e).__name__)
```

Select one:

- a. MyError ✓ The raised exception is of type MyError.
- b. Custom
- c. Error
- d. Exception

The correct answer is: MyError

Question 5

Correct

Mark 1.00 out of 1.00

```
What is the output?

import numpy as np

a = np.array([[1], [2], [3]])
b = np.array([10, 20, 30])
print((a + b).shape)
```

Select one:

- a. (3, 3) ✓
- b. (1, 3)
- c. (3, 1)
- d. (3,)

The correct answer is: (3, 3)

Question 6

Correct

Mark 1.00 out of 1.00

```
What is the output?

import pandas as pd

df = pd.DataFrame({
    'name': ['a', 'b', 'c'],
    'marks': [50, 80, 90]
})
print(df[df['marks'] > 70]['name'].tolist())
```

Select one:

- a. ['c']
- b. ['b', 'c'] ✓
- c. ['a', 'b']
- d. Error

The correct answer is: ['b', 'c']

Question 7

Correct

Mark 1.00 out of 1.00

```
What is the purpose of this code?

from scipy.optimize import minimize

f = lambda x: (x - 3)**2
res = minimize(f, x0=0)
print(round(res.x[0], 2))
```

Select one:

- a. Returns always 0
- b. Finds maximum near 0
- c. Finds minimum near 3 ✓
- d. Raises error

The correct answer is: Finds minimum near 3

Question 8

Correct

Mark 1.00 out of 1.00

```
What will this print?

import numpy as np

a = np.array([[1, 2], [3, 4]])
print(a.T[0][1])
```

Select one:

- a. 1
- b. 3 ✓
- c. 4
- d. 2

The correct answer is: 3

Question 9

Correct

Mark 1.00 out of 1.00

```
Which statement is true about this code?

import matplotlib.pyplot as plt
plt.hist([1,1,2,2,3,3], bins=3)
plt.show()
```

Select one:

- a. It creates a pie chart
- b. It creates 3 histogram bins ✓
- c. It raises an error because bins must be a list
- d. It creates 6 bars, one for each value

The correct answer is: It creates 3 histogram bins

Question 10

Correct

Mark 1.00 out of 1.00

```
What will this code do?

import pandas as pd

df = pd.DataFrame({
    'city': ['Mumbai', 'Pune', 'Mumbai'],
    'sales': [100, 200, 300]
})

print(df.groupby('city')['sales'].sum())
```

Select one:

- a. Sorts by city
- b. Counts rows by city
- c. Gives average sales
- d. Calculates total sales by city ✓

The correct answer is: Calculates total sales by city

Question 11

Correct

Mark 1.00 out of 1.00

```
What is the output?

import pandas as pd

df1 = pd.DataFrame({'id': [1, 2], 'name': ['A', 'B']})
df2 = pd.DataFrame({'id': [1, 3], 'marks': [90, 80]})

print(pd.merge(df1, df2, on='id'))
```

Select one:

- a. Error
- b. id name marks 0 1 A 90 ✓
- c. id marks 0 1 90 1 3 80
- d. id name 0 1 A 1 2 B

The correct answer is: id name marks 0 1 A 90

Question 12

Correct

Mark 1.00 out of 1.00

```
What is the output?

try:
    try:
        raise ValueError('bad')
    except ValueError as e:
        raise TypeError('worse') from e
except Exception as ex:
    print(type(ex).__name__)
    print(type(ex.__cause__).__name__)
```

Select one:

- a. ValueError then TypeError
- b. TypeError then NoneType
- c. TypeError then ValueError ✓
- d. Error

The correct answer is: TypeError then ValueError

Question 13

Correct

Mark 1.00 out of 1.00

```
What does this return?

import sqlite3

con = sqlite3.connect(':memory:')
cur = con.cursor()
cur.execute("CREATE TABLE emp(id INTEGER, sal INTEGER)")
cur.executemany("INSERT INTO emp VALUES(?, ?)", [(1, 1000), (2, 2000)])
cur.execute("SELECT AVG(sal) FROM emp")
print(cur.fetchone()[0])
```

Select one:

- a. 1000
- b. 2000
- c. Error
- d. 1500.0 ✓

The correct answer is: 1500.0

Question 14

Correct

Mark 1.00 out of 1.00

```
Why is commit() needed after insert/update/delete?
```

Select one:

- a. To print results
- b. To fetch records
- c. To close connection
- d. To save changes permanently ✓

The correct answer is: To save changes permanently

Question 15

Correct

Mark 1.00 out of 1.00

```
What is the output?

try:
    print(1 / 0)
except ZeroDivisionError:
    print("A")
finally:
    print("B")
```

Select one:

- a. A
- b. B
- c. A then B ✓ except runs first, then finally always executes.
- d. Error

The correct answer is: A then B

Question 16

Correct

Mark 1.00 out of 1.00

```
What will be the output?

import pandas as pd

df = pd.DataFrame({
    'A': [1, 2, 3],
    'B': [4, 5, 6]
})
print(df['A'].sum())
```

Select one:

- a. 6
- b. 3
- c. 5
- d. 15

The correct answer is: 6

Question 17

Correct

Mark 1.00 out of 1.00

```
What does this code visualize?

import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df = pd.DataFrame({
    'x': [1, 2, 3, 4],
    'y': [10, 20, 15, 25]
})

sns.scatterplot(data=df, x='x', y='y')
plt.show()
```

Select one:

- a. Histogram
- b. Heatmap
- c. Scatter plot
- d. Box plot

The correct answer is: Scatter plot

Question 18

Correct

Mark 1.00 out of 1.00

```
In mysql-connector-python, which placeholder is typically used in
parameterized queries?
```

Select one:

- a. {}
- b. :1
- c. ?
- d. %s ✓

The correct answer is: %s

Question 19

Correct

Mark 1.00 out of 1.00

```
What is the shape of the result?

import numpy as np

a = np.array([[1, 2], [3, 4], [5, 6]])
print(a[:, 1].shape)
```

Select one:

- a. (3, 1)
- b. (2,)
- c. (3,) ✓
- d. (1, 3)

The correct answer is: (3,)

Question 20

Incorrect

Mark 0.00 out of 1.00

```
Which statement is true about Plotly?
```

Select one:

- a. It supports interactivity like zoom and hover
- b. It is only for pie charts
- c. It only creates static images ✗
- d. It cannot be used in notebooks

The correct answer is: It supports interactivity like zoom and hover

Question 21

Correct

Mark 1.00 out of 1.00

```
What will be the output of this code?

def func():
    try:
        return 10
    finally:
        return 20

print(func())
```

Select one:

- a. 20 ✓ finally overrides the return from try.
- b. 10
- c. None
- d. Error

The correct answer is: 20

Question 22

Incorrect

Mark 0.00 out of 1.00

```
What is the issue in this code?

import sqlite3

con = sqlite3.connect("mydb.db")
cur = con.cursor()
cur.execute("CREATE TABLE student(id INTEGER PRIMARY KEY, name TEXT)")
cur.execute("INSERT INTO student VALUES(1, 'Rekha')")
cur.execute("INSERT INTO student VALUES(1, 'Asha')")
con.commit()
```

Select one:

- a. Primary key violation
- b. Name cannot be text
- c. Syntax error
- d. No issue

The correct answer is: Primary key violation

Question 23

Correct

Mark 1.00 out of 1.00

```
What will happen here?

try:
    x = int("12.5")
except ValueError:
    print("ValueError")
except Exception:
    print("Exception")
```

Select one:

- a. Prints ValueError int("12.5") raises ValueError.
- b. Prints Exception
- c. No output
- d. Program crashes

The correct answer is: Prints ValueError

Question 24

Incorrect

Mark 0.00 out of 1.00

```
What is the purpose of this code?

import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'A': [1, 2, 3],
    'B': [2, 4, 6]
})

sns.heatmap(df.corr(), annot=True)
plt.show()
```

Select one:

- a. Sorts dataframe
- b. Creates pair plot
- c. Plots missing values ⊗
- d. Shows correlation matrix

The correct answer is: Shows correlation matrix

Question 25

Incorrect

Mark 0.00 out of 1.00

```
What does a seaborn boxplot primarily help visualize?
```

Select one:

- a. Only category counts
- b. Only the mean
- c. Only pairwise correlation ⊗
- d. Distribution, quartiles, and outliers

The correct answer is: Distribution, quartiles, and outliers

Question 26

Correct

Mark 1.00 out of 1.00

```
What does this code create?

from plotnine import ggplot, aes, geom_point
import pandas as pd

df = pd.DataFrame({'x': [1, 2, 3], 'y': [4, 5, 6]})
p = ggplot(df, aes('x', 'y')) + geom_point()
print(p)
```

Select one:

- a. Pie chart
- b. Line chart
- c. Scatter plot ✓
- d. Histogram

The correct answer is: Scatter plot

Question 27

Incorrect

Mark 0.00 out of 1.00

```
What library is commonly used for ggplot-style plotting in Python?
```

Select one:

- a. matplotlib.gg ✗
- b. numpyplot
- c. pandasplot
- d. plotnine

The correct answer is: plotnine

Question 28

Correct

Mark 1.00 out of 1.00

```
What is the output?

import pandas as pd
import numpy as np

df = pd.DataFrame({
    'A': [1, 2, np.nan, 4]
})

df['A'].fillna(df['A'].mean(), inplace=True)
print(round(df['A'].sum(), 2))
```

Select one:

- a. 9.33 ✓ Mean of existing values is 2.33, so the sum becomes 9.33.
- b. 7.33
- c. 8.33
- d. Error

The correct answer is: 9.33

Question 29

Correct

Mark 1.00 out of 1.00

```
What is the output?

import numpy as np

a = np.array([1, 2, 3, 4])
print(a[1:3] * 2)
```

Select one:

- a. [2 4]
- b. [2 3]
- c. Error
- d. [4 6] ✓

The correct answer is: [4 6]

Question 30

Correct

Mark 1.00 out of 1.00

```
What is the output?

import pandas as pd

df = pd.DataFrame({'A':[10,20,30]}, index=['x','y','z'])
print(df.loc['y':'z', 'A'].sum())
```

Select one:

- a. 30
- b. 20
- c. 50 ✓
- d. Error

The correct answer is: 50

Question 31

Incorrect

Mark 0.00 out of 1.00

```
What is the effect of the color argument here?

import plotly.express as px
import pandas as pd

df = pd.DataFrame({'x':[1,2,3], 'y':[2,4,6], 'grp':['A','B','A']})
fig = px.scatter(df, x='x', y='y', color='grp')
```

Select one:

- a. Groups points by category using different colors
- b. Adds animation
- c. Changes marker size
- d. Turns the plot into a line chart ✗

The correct answer is: Groups points by category using different colors

Question 32

Correct

Mark 1.00 out of 1.00

```
What does this print?

from scipy import stats
import numpy as np

a = np.array([10, 20, 30])
print(np.round(stats.zscore(a).sum(), 5))
```

Select one:

- a. 1.0
- b. -1.0
- c. Error
- d. 0.0

The correct answer is: 0.0

Question 33

Incorrect

Mark 0.00 out of 1.00

```
What will be printed?

try:
    import numpy as np
    a = np.array([1, 2, 3])
    print(a / 0)
except Exception as e:
    print(type(e).__name__)
```

Select one:

- a. RuntimeWarning
- b. [inf inf inf]
- c. ZeroDivisionError
- d. Exception

The correct answer is: [inf inf inf]

Question 34

Incorrect

Mark 0.00 out of 1.00

```
In plotnine, which geom is commonly used for a bar chart?
```

Select one:

- a. geom_point() ✗
- b. geom_boxplot()
- c. geom_bar()
- d. geom_line()

The correct answer is: geom_bar()

Question 35

Correct

Mark 1.00 out of 1.00

```
What is the role of plt.show() here?

import matplotlib.pyplot as plt

plt.plot([1, 2, 3], [4, 5, 6])
plt.title("Line")
plt.show()
```

Select one:

- a. Adds grid
- b. Displays the graph ✓
- c. Saves figure
- d. Deletes the graph

The correct answer is: Displays the graph

Question 36

Correct

Mark 1.00 out of 1.00

```
What will be the output?

import pandas as pd

df = pd.DataFrame({'A':[1,1,2,2], 'B':[3,3,4,5]})
print(df.duplicated().sum())
```

Select one:

- a. 2
- b. 1 ✓
- c. 0
- d. 3

The correct answer is: 1

Question 37

Correct

Mark 1.00 out of 1.00

```
Which graph is produced?

import matplotlib.pyplot as plt

plt.bar(['A', 'B', 'C'], [10, 20, 15])
plt.show()
```

Select one:

- a. Pie chart
- b. Bar chart ✓
- c. Scatter plot
- d. Histogram

The correct answer is: Bar chart

Question 38

Correct

Mark 1.00 out of 1.00

```
What happens here?

import pandas as pd

df = pd.DataFrame({
    'A': [1, None, 3]
})
print(df['A'].mean())
```

Select one:

- a. Error
- b. 1.33
- c. 2.0 ✓
- d. None

The correct answer is: 2.0

Question 39

Correct

Mark 1.00 out of 1.00

```
What does this return?

import pandas as pd

df = pd.DataFrame({
    'A': ['1', '2', '3']
})
df['A'] = df['A'].astype(int)
print(df['A'].dtype)
```

Select one:

- a. object
- b. string
- c. int64 ✓
- d. float64

The correct answer is: int64

Question 40

Incorrect

Mark 0.00 out of 1.00

```
What does this code compute?

from scipy.integrate import quad

result, _ = quad(lambda x: x**2, 0, 2)
print(result)
```

Select one:

- a. 2 ✗
- b. 6
- c. 8/3
- d. 4

The correct answer is: 8/3